

DEMO

[Snowflake for Developers](#) / [Guides](#) / [Liquidity Risk Agent: AI-Powered LCR Forec...](#)

Liquidity Risk Agent: AI-Powered LCR Forecasting with Snowflake



Cortex Analyst



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[FORK REPO](#)



sfguide-liquidity-risk-agent

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Liquidity Risk Agent

Basel III Liquidity Coverage Ratio (LCR) Analysis on Snowflake

An end-to-end solution for 150-day LCR forecasting, stress-test what-if scenarios, and AI-powered natural language analytics -- built entirely within Snowflake.

how financial institutions can transform their liquidity risk workflows by integrating structured position data with intelligent analytics...

The Challenge

After the 2008 financial crisis, Basel III introduced the Liquidity Coverage Ratio (LCR) to ensure banks hold enough high-quality liquid assets (HQLA) to survive 30 days of cash outflows under stress.

$$\text{LCR} = \text{HQLA} / \text{Net Cash Outflows (30days)} \geq 100\%$$



Fragmented Tooling

Calculations in spreadsheets,
data in warehouses, reports
in separate BI tools



Slow Scenario Analysis

Days to run what-if stress
tests across multiple
systems and teams



Limited Self-Service

Risk analysts depend on
data engineers for every
ad-hoc liquidity question

Solution Overview

Everything runs inside Snowflake -- no external infrastructure, no data movement.



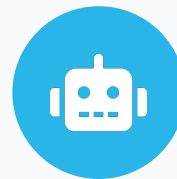
LCR Dashboard

Real-time 150-day LCR
forecast with HQLA &
net cash outflow metrics.
One-click recalculation
via Snowflake Notebooks.



What-If Scenarios

Stress-test with pre-defined
scenarios. Zero-copy cloning
isolates sandbox data.
Multiplicative factor
application across tables.



AI Agent

Natural language questions
answered via Cortex
Analyst + Semantic Views.
Auto-generated SQL with
smart chart rendering.

Business Impact

Transforming liquidity risk management with a unified Snowflake-native solution.



Regulatory Compliance

Ensure Basel III LCR requirements are met with accurate, auditable calculations and 150-day forward projections that identify potential compliance breaches before they occur.



Proactive Risk Management

Identify days where LCR may fall below the 100% threshold and take corrective action through what-if scenario analysis to understand the impact of potential stress events.



Operational Efficiency

Replace manual spreadsheet-based calculations with automated, on-demand LCR computation that runs in seconds and provides consistent, reproducible results.



Faster Decision Making

Treasury teams get immediate answers to complex liquidity questions through natural language queries, eliminating the need to wait for data teams or build custom reports.



Scenario Planning

Stress test the impact of market events, counterparty failures, and regional crises before they happen, enabling proactive liquidity buffer management.



Simplified Infrastructure

Eliminate external data movement, separate BI tools, and manual data pipelines. Everything runs natively within Snowflake.

Architecture

SNOWFLAKE PLATFORM

PRESENTATION



Streamlit
Dashboard



What-If
Scenarios



Cortex Agent
Chat



Snowflake
Intelligence

COMPUTE



Notebook:
LIQUIDITY_FORECAST
Baseline LCR



Notebook:
WHAT_IF_FORECAST
Stress Tests



Cortex Analyst
+ Semantic View
LIQUIDITY_SV

Snowpark
Python

DATA



RAW Schema

Positions (580K) | Inflows (100K)
Outflows (100K) | Market Data (3.6M)
Counterparties | Asset Classifications

CLONE



RAW_SANDBOX

Zero-copy clones for
what-if isolation



PRESENTATION

LCR results (150 days)
What-If LCR results

Data Model

Reference Tables

Table	Description
ASSET_CLASSIFICATIONS	13 security types, HQLA levels, haircuts
BUSINESS_UNIT_REFERENCE	9 business units (US/EU/Asia)
COUNTERPARTY_DATA	2,000 counterparties
CURRENCY_REFERENCE	10 major currencies + FX rates
STRESS_SCENARIOS	3 Basel III scenarios
WHAT_IF_DEFINITIONS	Pre-defined stress factors

Fact Tables

Table	Description
POSITIONS	~580K rows (500K securities + 80K cash)
CASH_INFLOWS	100K rows (deposits, loan repayments)
CASH_OUTFLOWS	100K rows (withdrawals, payments)
MARKET_DATA	~3.65M rows (10K sec x 365 days)
LCR	150-day LCR forecast results
WHAT_IF_LCR	Scenario comparison results

All synthetic data is generated in-place using Snowflake's TABLE(GENERATOR(...)) -- no external data loading required.

580K Positions

200K Cash Flows

3.65M Market Data Points

2K Counterparties

LCR Calculation Pipeline

The 150-day forecast is computed via Snowpark Python in Snowflake Notebooks, called from Streamlit via EXECUTE NOTEBOOK.



What-If Scenario Engine

1

Clone Data

Zero-copy clone of Positions, Inflows, Outflows into sandbox. No data duplication cost.

>

2

Look Up Factors

Query WHAT_IF_DEFINITIONS for scenario-specific stress factors (region, type, etc.).

>

3

Apply Factors

MERGE updates sandbox tables. Multiplicative via REDUCE(ARRAY_AGG(FACTOR)).

>

4

Recalculate LCR

Run full LCR pipeline on stressed sandbox data. Results to WHAT_IF_LCR.

Example Scenario #1

Target	Dimension	Factor
European Counterparties	Region = Europe	-10%
Institutional Counterparties	Type = Institutional	-15%
Government Bonds AAA	Security Type	-5%
Main Index Equities	Security Type	-10%
Treasury Business Units	BU Type = Treasury	-5%

Cortex Agent & Semantic View

How It Works



User asks a question in natural language via the Streamlit chat interface



Cortex Analyst interprets the question against the LIQUIDITY_SV semantic view



SQL is auto-generated from the semantic model (9 tables, 50+ dimensions/facts)



Results are rendered with smart chart detection (line, pie, bar based on columns)

LIQUIDITY_SV Semantic View



9 base tables with full relationships



Covers: Positions, Inflows, Outflows, Counterparties, Business Units, Assets, LCR results, What-If results



Dimensions: Region, BU Type, Security Type, HQLA Level, etc.



Facts: Position Value, Flow Amounts, LCR Ratio, HQLA totals, Haircuts



Verified queries for common patterns



Registered in Snowflake Intelligence for enterprise-wide access

Snowflake Features Used



Snowflake Notebooks

EXECUTE NOTEBOOK for
programmatic LCR calculations



Snowpark Python

DataFrame API for HQLA
projection & cashflow netting



Streamlit in Snowflake

3-page interactive app
running natively in Snowflake



Cortex Analyst

Text-to-SQL via semantic
view for ad-hoc analytics



Cortex Agent

LIQUIDITY_FORECAST_AGENT
with analyst tool



Snowflake Intelligence

Enterprise-wide agent
access & discovery



Zero-Copy Cloning

Instant sandbox creation
for what-if isolation



Semantic Views

LIQUIDITY_SV with 9 tables,
relationships & verified queries

User Experience

LCR Dashboard

- ✓ Today's LCR, HQLA, and net cash outflows as metric cards
- ✓ 150-day LCR trend line chart with compliance threshold at 1.0
- ✓ One-click recalculation runs EXECUTE NOTEBOOK in real-time
- ✓ Raw data expandable table view

What-If Scenarios

- ✓ Dropdown to select pre-defined stress scenario IDs
- ✓ View scenario definition before execution (factors & targets)
- ✓ Execute clones data, applies stress factors, and recalculates LCR
- ✓ Side-by-side comparison charts

Ask the Agent

- ✓ Chat interface with sample questions for quick-start
- ✓ Natural language to SQL via Cortex Analyst API
- ✓ Auto-rendered charts: line for LCR, pie for HQLA, bar for BU breakdowns
- ✓ Conversation history with clear chat

Getting Started

Single SQL script sets up the entire environment -- role, warehouse, database, schemas, tables, synthetic data, notebooks, Streamlit app, semantic view, and agent.

1

Run setup.sql (Sections 1-9)

Creates role, warehouse, database, schemas, tables, reference data, and generates 800K+ synthetic rows

2

Upload files to stages

PUT notebooks and Streamlit app files to internal stages (LIQUIDITY_NOTEBOOK_STAGE, LIQUIDITY_STREAMLIT_STAGE)

3

Run setup.sql (Sections 11-13)

Creates Snowflake Notebooks, Streamlit app, Semantic View (LIQUIDITY_SV), Cortex Agent, and Snowflake Intelligence

4

Open Streamlit app

Navigate to LIQUIDITY_STREAMLIT in Snowsight. Click 'Re-Calculate LCR' to run the first forecast.



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Prerequisites: Snowflake account with ACCOUNTADMIN (for initial setup), Medium warehouse. Teardown script included for cleanup.

Thank You

From spreadsheet chaos to AI-powered liquidity intelligence --
all within Snowflake's unified platform.



Basel III
Compliant



Zero Data
Movement



AI-Powered
Analytics



Self-Service
Scenarios

quickstarts.snowflake.com/guide/liquidity-risk-agent